A good forecast can change with the evidence

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Revision / ES: revisión / PT: revisão

Name the change and show its effect.

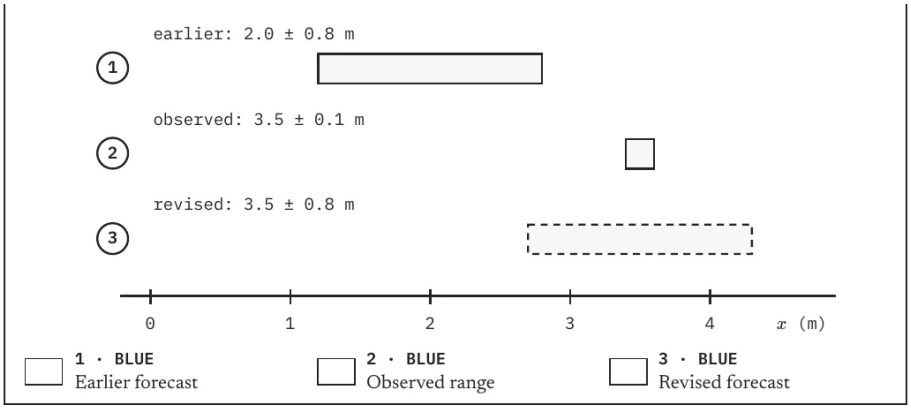
Also on your note page: Prediction, Evidence, Limit.

B READ THE RELATIONSHIP

$x_f = x_0 + vt$. With exact time, half-width = starting bound + t × velocity bound.

As you read, find: claim · evidence · limit.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Keep $x_0 = 8 \pm 0.5$ m and exact $t = 3$ s . A new measurement changes v from −2.0 to −1.5 ± 0.1 m/s . Update the forecast.

Given $x_0 = 8 \pm 0.5$ m $v = -1.5 \pm 0.1$ m/s exact $t = 3$ s

Rule Update center and conservative bound.

Work $x_f = 8 - 1.5(3)$ $\delta x = .5 + 3(.1) = 0.8$ m

Answer 2.7 to 4.3 m Center shifts +1.5 m .

E SAVE FOR THE FINAL MEETING

Show the evidence behind XJ's forecast and what changed after review. Separate the physics from cost and feasibility decisions.

Read the physics: Revision

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Give a predicted position, a time and a reference frame. Show the measurement and calculation that support your answer. List any assumptions.

02 HOW IT WORKS

A model may give a central position and a range. A bound says which values the model allows; it does not give the chance of an impact. Compare the old and new results on one scale.

03 WHAT TO CHECK

Name the step that changed. Explain what it did to the forecast and what the physics still cannot decide.

04 TRY THE EXAMPLE FROM YOUR NOTES

Keep x 0 = 8 ± 0.5 m and exact t = 3 s . A new measurement changes v from −2.0 to −1.5 ± 0.1 m/s . Update the forecast. 2.7 to 4.3 m Center shifts +1.5 m .

Model check: $x_f = x_0 + vt$. With exact time, half-width = starting bound + t × velocity bound.

05 CONNECT TO XJ

Show the evidence behind XJ's forecast and what changed after review. Separate the physics from cost and feasibility decisions.

SAY IT BEFORE YOU WRITE

Which piece of evidence would make you revise again? We revised ___ after ___. Our forecast still assumes ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Revision / ES: revisión / PT: revisão

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, revision means …	A new sighting changes your forecast. Which result changes, and what new evidence explains it?
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Doodle two forecast bands on the same scale. Add a note connecting the revision to a specific measurement.

Labels you can use: claim · evidence · limit.

Choose zero, direction, and scale.



Tell a partner: We revised ___ after ___. Our forecast still assumes ___.

On assignment: Revision

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two short-interval forecasts in the same frame give 4–6 km and 5–7 km. The new estimate uses a later sighting. A headline says "Impact confirmed."

PICK A JOB / DELIVER ONE PRODUCT

☐ **Forecast engineer**

Overlay the two bands. Issue a revised location bulletin and name what observation would trigger another update.

☐ **Science news editor**

Decide whether to air the headline. Build a graphic with both intervals and write the replacement line you will use.

Word bank: claim · evidence · limit.

Try: "Our desk recommends ___. This mark shows ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

"We first said ___. Now we ___ because ___."

Editor's check: Separate the physics forecast from a decision about action.